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REVIEW

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Explainable AI for differential diagnosis of skin-manifesting neglected tropical diseases (NTDS) in darker skin tones

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Abstract

Background Skin-manifesting neglected tropical diseases (NTDs) pose significant diagnostic challenges due to overlapping clinical presentations and limited access to specialist care in endemic regions. Artificial intelligence (AI) has shown promise in dermatological diagnosis; however, concerns remain regarding algorithmic bias, reduced accuracy in darker skin tones, and lack of transparency in decision-making.

Aim This review aimed to synthesise existing evidence on explainable AI approaches for the differential diagnosis of skin-manifesting NTDs, with emphasis on performance, equity across dark skin tones, and clinical applicability.

Methods A structured narrative review was conducted using systematic search methods across PubMed/MEDLINE, Scopus, AJOL, and ScienceDirect. Eligible studies included peer-reviewed AI-based diagnostic research involving skin conditions that incorporated explainability or interpretability methods. Literature published between 2015 and 2025 was screened and synthesised thematically.

Results Evidence from studies demonstrated that deep learning models achieve high diagnostic performance in dermatology (often > 85% accuracy), but consistently underperform in darker skin tones, with reported reductions of up to 20%. Explainable AI techniques such as saliency maps, Grad-CAM, and confidence scoring were shown to enhance interpretability and support differential diagnosis, though limitations related to dataset diversity and real-world deployment persist.

Conclusion Explainable AI represents a critical advancement for equitable and reliable diagnosis of skin-manifesting NTDs. Addressing dataset bias, embedding transparency, and aligning AI tools with frontline workflows are essential to maximise clinical and public health impact.

Clinical trial number Not applicable.

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Keywords Neglected tropical diseases, Explainable artificial intelligence, Skin of colour, Differential diagnosis, Dermatology AI, Global health equity

Introduction

Neglected tropical diseases (NTDs) constitute a group of communicable diseases that disproportionately affect impoverished populations in tropical and subtropical regions, contributing to significant morbidity, disability, and socioeconomic burden [1–4]. A substantial subset of these conditions manifests primarily through cutaneous lesions, including scabies, yaws, cutaneous leishmaniasis, Buruli ulcer and leprosy, which often present with overlapping visual features that complicate clinical diagnosis, particularly in primary care and resource-limited settings [1, 5]. The intersection of high disease burden and diagnostic complexity imposes considerable challenges that traditional clinical workflows struggle to address efficiently.

In recent years, artificial intelligence (AI) has emerged as a transformative tool for medical image analysis, achieving notable performance in dermatological applications such as skin cancer detection and lesion classification [6–9]. AI techniques, particularly deep learning, can automatically identify salient features from clinical images without explicit programming, offering scalable diagnostic support where experts are scarce [9]. However, a major limitation of many existing AI models is their development on datasets dominated by lighter skin tones, which introduces algorithmic bias and significantly degrades performance when applied to individuals with darker skin [10, 11]. In this review, "darker skin tones" refer to Fitzpatrick skin types IV–VI, a standard dermatological classification based on melanin content and response to ultraviolet exposure [11]. Moreover, existing studies in the NTD domain remain sparse, with only a few pilot investigations reporting moderate classification accuracy (≈70%) for multiple skin NTDs using convolutional neural networks trained on limited images from African populations [12], underscoring the nascent state of this field.

Beyond raw predictive performance, the interpretability of AI decisions has become an ethical and practical demand in clinical settings. Traditional "black-box" models provide little insight into *how* a decision was reached, making clinicians reluctant to trust or adopt these systems without transparent reasoning [12]. Explainable AI (XAI) techniques, such as saliency maps and attention heatmaps, aim to make model reasoning interpretable by highlighting features that contribute most to diagnostic predictions, thereby enabling clinicians to corroborate algorithmic suggestions with clinical understanding [12–14]. Critically, the integration of explainability into AI for differential diagnosis the process of distinguishing between diseases with similar features could enhance diagnostic confidence, reduce misclassification, and help tailor management decisions in skin-manifesting NTDs, particularly in dark-skinned populations historically neglected in both AI research and dermatological datasets.Resolved="yes"-->

Despite these advances, no comprehensive review has synthesised the current evidence on explainable AI approaches specifically focused on skin NTD diagnosis in darker skin tones, and how such methods might ameliorate diagnostic disparities and support frontline care. Therefore, this structured narrative review aims to critically examine and synthesise existing literature on explainable AI methodologies applied to differential diagnosis of skin-manifesting NTDs, with particular emphasis on performance, bias, and applicability in darker skin populations.

Methods

Review design

This structured narrative review employed systematic search methods across multiple databases to identify relevant literature and this narrative approach was considered most appropriate given the emerging and interdisciplinary nature of XAI applications in the diagnosis of skin-manifesting NTDs. Also, preprints and WHO reports were used as contextual and supporting literature, particularly in an emerging field where peer-reviewed evidence remains limited. Where reported, studies were assessed for representation and performance across Fitzpatrick skin types (IV–VI) as an indicator of darker skin tone inclusion and potential bias.

Literature search strategy

Search was conducted to identify studies examining AI-based diagnostic approaches relevant to skin-manifesting NTDs and skin of colour. Searches were performed across four major bibliographic databases: PubMed/MEDLINE, Scopus, Web of Science, and IEEE Xplore, selected to capture both biomedical and computational research outputs. The search timeframe was restricted from January 2015 to December 2025, reflecting the contemporary period of accelerated deep learning and explainable AI development.

Search terms were developed iteratively and combined using Boolean operators, structured around four core conceptual domains:

1. Skin-manifesting NTDs (e.g., scabies, yaws, leprosy, cutaneous leishmaniasis),

Table 1 Database-specific search strategies for identifying studies on explainable AI in skin-manifesting NTDs (2015–2025)

Database	Timeframe	Search Strategy (Full Boolean Query)	Filters Applied	Records Retrieved
PubMed/MEDLINE	Jan 2015 – Dec 2025	("neglected tropical diseases" OR "skin NTDs" OR scabies OR yaws OR leprosy OR "cutaneous leishmaniasis" OR "Buruli ulcer") AND ("artificial intelligence" OR "machine learning" OR "deep learning" OR "convolutional neural network" OR CNN) AND ("explainable AI" OR "interpretable AI" OR "model explainability" OR Grad-CAM OR "attention map") AND ("skin of colour" OR "dark skin" OR "darker skin tones" OR "melanin-rich skin")	English; Human; Full-text	4,213
Scopus	Jan 2015 – Dec 2025	TITLE-ABS-KEY(("skin" AND NTD*) AND ("artificial intelligence" OR "deep learning") AND (explain* OR interpretab*) AND ("skin of colour" OR dark*))	Articles; Reviews; English	1023
Web of Science	Jan 2015 – Dec 2025	TS=(("skin" AND "infectious disease*") AND ("artificial intelligence") AND (explainable OR interpretable))	SCI-Expanded; English	6,232
IEEE Xplore	Jan 2015 – Dec 2025	("medical image analysis" AND "explainable AI") OR ("deep learning" AND dermatology AND explainability)	Journals; Conferences	2,342
Manual Search	—	Reference list screening and backward citation tracking of key studies and WHO reports	Relevant sources only	121

Note: Boolean operators ("AND", "OR") were adapted for database-specific syntax. Searches were iteratively refined to balance sensitivity and specificity

Table 2 Study selection criteria (guiding framework)

Component	Inclusion Criteria	Exclusion Criteria
Study type	Peer-reviewed original studies and relevant reviews; selected high-quality grey literature for contextual insight	Editorials, opinion pieces without analytical relevance, conference abstracts without full data
Population / Focus	Studies on AI in dermatology, infectious diseases, or related diagnostic imaging	Studies unrelated to diagnostic AI or not involving medical/dermatological conditions
Intervention	Use of artificial intelligence, machine learning, or deep learning models with explainability approaches	Studies without AI or without relevance to explainability/interpretability
Outcomes	Explainability methods, trust, bias, fairness, clinical relevance	Studies not addressing interpretability, bias, or clinical implications
Context	Skin-manifesting diseases, dermatological imaging, or adjacent conditions informing bias in skin tone	Non-healthcare AI applications
Language	English	Non-English publications
Publication type	Peer-reviewed articles; selected grey literature (e.g., WHO reports) for background/context only	Preprints used as primary evidence

2. Artificial intelligence and deep learning,
3. Explainability and interpretability, and.
4. Skin of colour or darker skin tones.

Synonyms and controlled vocabulary (e.g., MeSH terms where applicable) were incorporated to maximise sensitivity. A detailed search strategy table (Table 1) documents database-specific queries, keywords, Boolean combinations, and filters applied, ensuring reproducibility of the search process.

Eligibility criteria

Study eligibility was determined using predefined inclusion and exclusion criteria to ensure relevance and methodological rigour. Inclusion criteria comprised peer-reviewed studies and that:

(i) Employed AI-based or machine-learning diagnostic methods involving skin images or cutaneous conditions;
(ii) Explicitly addressed explainability, interpretability, or transparency of AI model outputs (e.g., saliency maps, Grad-CAM, attention mechanisms); and.
(iii) Focused on skin-manifesting NTDs or clinically similar infectious dermatoses with relevance to endemic or low-resource settings.

Exclusion criteria included:

(i) Non-AI or purely rule-based diagnostic studies;
(ii) Studies limited to cosmetic or aesthetic dermatology without public health relevance; and.

A comprehensive eligibility table (Table 2) summarises inclusion and exclusion criteria, alongside their methodological justification.

Synthesis

Given the conceptual and methodological diversity of the included studies, findings were synthesised narratively

across thematic domains rather than statistically pooled. This approach enabled critical comparison of how explainable AI has been operationalised, where biases persist, and how model transparency influences clinical applicability in darker skin populations. The synthesis emphasised cross-cutting themes, methodological gaps, and implications for equitable AI deployment in skin-related NTD diagnosis. The characteristics of the studies is shown in Table 3 below.

Findings

Overview of skin-manifesting NTDs relevant to AI diagnosis

Skin-manifesting NTDs represent a clinically heterogeneous group of infections whose primary diagnostic features are visual, yet frequently ambiguous. Conditions such as scabies, yaws, leprosy, cutaneous leishmaniasis and Buruli ulcer account for a substantial proportion of dermatological morbidity in endemic regions, particularly across sub-Saharan Africa, South-East Asia and parts of Latin America [1]. These diseases commonly present with erythema, nodules, ulcers, plaques or papular eruptions, features that are neither pathognomonic nor disease-specific, thereby creating significant diagnostic uncertainty in routine clinical practice [5, 12].

The diagnostic challenge is further compounded by visual overlap with non-NTD conditions, including fungal infections, bacterial pyodermas, eczema, psoriasis and insect-bite reactions, which frequently coexist in the same ecological and socioeconomic settings [1]. For example, scabies may mimic atopic dermatitis or impetigo, while early leprosy lesions are often mistaken for tinea or vitiligo, particularly in individuals with darker skin tones where erythema and inflammation are less conspicuous [12]. Cutaneous leishmaniasis and Buruli ulcer may both present as chronic, painless ulcers, necessitating laboratory confirmation that is often unavailable in primary-care or rural settings [1].

The visually driven nature of these diseases makes them theoretically well suited for image-based artificial intelligence (AI) diagnostics, particularly using deep learning models capable of detecting subtle morphological patterns beyond human perception [9]. Unlike laboratory-dependent diagnostics, AI systems trained on clinical images offer scalable decision support that can be deployed on smartphones or portable devices, addressing workforce shortages in endemic regions [12]. However, the effectiveness of such systems is contingent on their ability to distinguish between clinically similar conditions and to generalise across diverse skin tones, a limitation

Table 3 Characteristics of included studies on explainable AI in dermatological and skin-related diagnostics

Study (Author, Year)	Study Design	Disease Focus	Dataset / Sample Size	Skin Tone Representation	AI Model Used	Explainability Method	Key Findings	Bias / Limitations
Esteva et al. (2017)	Experimental (CNN validation study)	Skin cancer (melanoma vs. benign lesions)	129,450 images	Predominantly light skin tones	Convolutional Neural Network (CNN)	Limited (post-hoc interpretability)	Achieved dermatologist-level accuracy (~91%)	Lack of diversity; poor representation of darker skin
Han et al. (2018)	Retrospective diagnostic study	Onychomycosis (fungal nail infection)	49,567 images	Not stratified	CNN (region-based)	Not explicitly reported	Comparable or superior to dermatologists	No reporting on bias or skin tone performance
Groh et al. (2021)	Observational evaluation study	General dermatological conditions	Fitzpatrick17k dataset	Explicit (Fitzpatrick I–VI)	Deep neural networks	Saliency-based evaluation	Lower performance in darker skin tones	Dataset imbalance; subgroup disparities
Daneshjou et al. (2021)	Analytical fairness study	Multiple dermatological diseases	Diverse clinical datasets	Stratified (skin tone groups)	Deep learning models	Grad-CAM, attribution methods	Accuracy reduced by up to 15% in darker skin	Algorithmic bias; poor generalisation
Du et al. (2022)	Experimental bias mitigation study	Dermatology AI fairness	Not specified	Skin tone groups included	Disentanglement-based deep learning	Implicit interpretability	Improved fairness after bias mitigation	Baseline models showed strong bias
Munia & Imran (2025)	Experimental generative modelling study	Dermatology AI bias mitigation	Synthetic + real datasets	Racial/skin tone groups	Diffusion models	Not explicitly XAI-focused	Improved performance in darker skin tones	Synthetic data reliance; validation needed
Singh et al. (2024)	Applied AI diagnostic study	Multiple skin diseases	Not specified	Limited reporting	CNN-based model	Grad-CAM	Improved interpretability and diagnostic transparency	Dataset limitations; unclear generalisability
Minyilu et al. (2025)	Systematic review	Skin NTDs	Multiple datasets	Not consistently reported	Multimodal deep learning	Not consistently applied	Moderate diagnostic accuracy (~70%) in NTDs	Sparse data; lack of XAI integration

Table 4 Common skin-manifesting NTDs, key visual features, and diagnostic challenges

Disease	Typical Clinical Presentation	Common Mimickers	Key Diagnostic Challenges	Relevance for Image-Based AI
Scabies	Generalised pruritic papules, burrows, excoriations	Eczema, impetigo, insect bites	Non-specific lesions; secondary infection	Pattern recognition of burrows and lesion distribution
Yaws	Papillomatous or ulcerative skin lesions	Tropical ulcers, bacterial pyoderma	Overlap with non-treponemal ulcers	Lesion morphology and progression tracking
Leprosy	Hypopigmented or erythematous plaques, nodules	Tinea, vitiligo, psoriasis	Subtle early lesions; sensory loss under-detected	Detection of lesion borders and texture changes
Cutaneous leishmaniasis	Chronic ulcerative or nodular lesions	Buruli ulcer, skin cancer	Requires parasitological confirmation	Ulcer edge and granulation pattern analysis
Buruli ulcer	Painless nodules progressing to ulcers	Leishmaniasis, bacterial ulcers	Late presentation; limited diagnostics	Temporal lesion evolution recognition



Fig. 1 Conceptual diagnostic pipeline for skin image analysis

that remains inadequately addressed in current AI dermatology research [10]. Consequently, skin-manifesting NTDs present a compelling yet under-explored use case for explainable AI, where transparency in feature attribution is essential for safe differential diagnosis and clinical trust (Table 4).

Current AI applications in skin disease diagnosis

Convolutional neural networks (CNNs) have become the dominant architecture in skin disease classification due to their capacity to hierarchically extract discriminative visual features such as colour gradients, texture irregularities and lesion boundaries directly from raw images [15, 16]. These findings have catalysed widespread interest in extending AI-driven diagnostics beyond oncology to infectious and inflammatory dermatoses, including conditions prevalent in low- and middle-income countries.

Recent advances have further enabled the deployment of smartphone-based and image-centric AI tools, reflecting a shift toward decentralised and scalable diagnostic support [2, 3, 8]. Mobile devices equipped with embedded or cloud-based AI models allow frontline health workers to capture clinical images and receive near-real-time diagnostic suggestions, thereby addressing critical shortages of dermatology specialists in endemic settings [1, 12]. Such systems have shown promise in pilot studies targeting scabies, fungal infections and other visually driven skin diseases, achieving moderate to high classification accuracy even under variable image quality conditions [12]. The conceptual AI diagnostic pipeline underpinning these applications spanning image acquisition, preprocessing, feature extraction, classification and output interpretation is illustrated in Fig. 1.

Despite these advances, existing AI dermatology models exhibit structural limitations that constrain

their clinical utility, particularly for skin-manifesting NTDs. Most systems are trained on datasets dominated by lighter skin tones and non-infectious conditions, resulting in reduced generalisability and increased misclassification in darker-skinned populations [10, 11]. Furthermore, many high-performing models operate as black boxes, providing probabilistic outputs without transparent justification, which undermines clinician trust and limits adoption in safety-critical contexts [12]. These limitations are especially problematic for differential diagnosis, where subtle visual distinctions determine treatment pathways.

Gaps in AI performance across skin tones

Growing empirical evidence demonstrates that many AI systems used in dermatological diagnosis exhibit systematically reduced performance in individuals with darker skin tones, raising concerns about both clinical reliability and equity [4, 18–20]. Large-scale evaluations of dermatology datasets have shown that classification accuracy, sensitivity and lesion localisation performance are consistently lower for Fitzpatrick skin types V–VI compared with types I–III, with reported accuracy differentials ranging from 5% to over 20% depending on task complexity and disease category [21–23]. These disparities are particularly pronounced for inflammatory and infectious dermatoses, where colour contrast and erythema are less visually salient in darker skin, increasing the likelihood of misclassification [12].

At the core of this performance gap lies dataset imbalance and algorithmic bias. Most publicly available dermatology image repositories have been constructed using data from high-income settings and predominantly light-skinned populations, resulting in skewed training distributions that fail to capture the phenotypic variability of skin disease across pigmentation spectra [11, 22]. Deep learning models trained on such datasets internalise these biases, optimising performance for over-represented groups while generalising poorly to under-represented skin tones [10]. Importantly, performance metrics reported in many AI studies are aggregated across populations, masking subgroup-specific errors and obscuring clinically meaningful disparities [23, 24]. The lack of routine stratification by skin tone in model evaluation further perpetuates a false perception of generalizability (Table 5).

The clinical and ethical implications of these gaps are substantial, particularly in the context of skin-manifesting NTDs that disproportionately affect populations with darker skin. Diagnostic inaccuracies may lead to delayed treatment, inappropriate therapy, or missed opportunities for early intervention, thereby exacerbating disease burden and undermining NTD control efforts [1].

Explainable AI techniques used in medical image analysis

It is important to note that XAI techniques have emerged as a critical response to the opacity of deep learning models in medical image analysis, particularly in high-stakes diagnostic contexts. In dermatology and infectious disease imaging, XAI seeks to bridge the gap between algorithmic predictions and clinical reasoning by making model decision processes interpretable to end users [25]. Saliency maps have been applied across dermatological tasks to highlight lesion boundaries, colour heterogeneity and textural features, thereby offering clinicians a visual rationale for AI outputs. However, their clinical utility remains contested, as saliency outputs can be noisy, unstable, and sensitive to minor perturbations in input images, limiting their reliability for diagnostic decision-making [26].

To address these limitations, more robust techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) and attention-based heatmaps have gained prominence. Grad-CAM localises discriminative regions by weighting convolutional feature maps according to their contribution to a given class prediction, producing spatially coherent heatmaps that are more intuitively interpretable than raw saliency gradients [27, 28]. In dermatology, Grad-CAM has been shown to improve clinician trust by demonstrating whether models focus on clinically relevant lesion areas rather than background artefacts [12, 16]. Attention mechanisms further extend

Table 5 Summary of AI dermatology studies reporting performance across skin tones

Study	AI Task	Skin Tone Stratification	Key Findings	Reported Bias or Limitation
Adamson & Smith [21]	Skin cancer classification	Fitzpatrick I–VI	Lower sensitivity in darker skin tones	Under-representation of dark skin images
Daneshjou et al. [22]	Multiple dermatological diagnoses	Fitzpatrick I–VI	Accuracy decreased by up to 15% in types V–VI	Dataset imbalance
Groh et al. [23]	Dermatology image classification	Light vs. dark skin	Aggregated metrics masked subgroup errors	Lack of stratified reporting
Du et al. [11]	Fairness in dermatology AI	Skin tone groups	Bias reduced with disentanglement methods	Baseline models showed strong bias
Munia & Imran [10]	Bias mitigation using diffusion models	Racial and skin tone groups	Improved performance in darker skin	Highlights bias in standard CNNs

this paradigm by enabling models to learn where to "attend" within an image during training, thereby embedding interpretability into the model architecture itself rather than applying it post hoc [29].

In addition, beyond spatial explanations, confidence scoring and decision visualisation play a complementary role in explainability (Fig. 2). Probability estimates, uncertainty quantification and confidence thresholds allow clinicians to assess the reliability of AI outputs and identify cases requiring referral or further investigation [30]. This is particularly relevant in differential diagnosis of visually overlapping conditions, where overconfident yet incorrect predictions may carry significant clinical risk. Nevertheless, confidence scores are often poorly calibrated, and high confidence does not necessarily equate to correctness, especially in under-represented populations [22]. Also, while explainable AI techniques enhance transparency and foster clinical engagement, their interpretability remains context-dependent rather than absolute.

Differential diagnosis: distinguishing clinically similar skin NTDs

Differential diagnosis remains one of the most complex challenges in the clinical management of skin-manifesting NTDs, owing to substantial overlap in morphological features across conditions. Diseases such as scabies, yaws, leprosy, cutaneous leishmaniasis and Buruli ulcer frequently present with non-specific lesions, including papules, nodules, plaques and chronic ulcers, which evolve dynamically over time and are influenced by host immune response and secondary infection [1, 5]. In endemic settings, where laboratory confirmation is limited, clinicians often rely on visual assessment alone, increasing the likelihood of diagnostic error, particularly during early or atypical disease stages [12].

From an AI perspective, this diagnostic complexity underscores the importance of multi-class classification frameworks rather than binary disease–non-disease models. Binary classifiers may achieve high apparent accuracy yet fail in real-world settings where multiple clinically similar conditions coexist, leading to unsafe oversimplification of diagnostic pathways [15, 29]. Multi-class AI models are better aligned with clinical reasoning, enabling simultaneous differentiation among several plausible diagnoses and supporting more realistic decision-making processes [16]. However, multi-class classification also introduces greater susceptibility to misclassification, particularly when training data are imbalanced or phenotypically homogeneous.

The risk of false positives and false negatives in AI-assisted differential diagnosis carries significant clinical consequences. False positives may result in unnecessary antiparasitic or antibiotic treatment, contributing to adverse effects and antimicrobial resistance, while false negatives can delay treatment and increase transmission

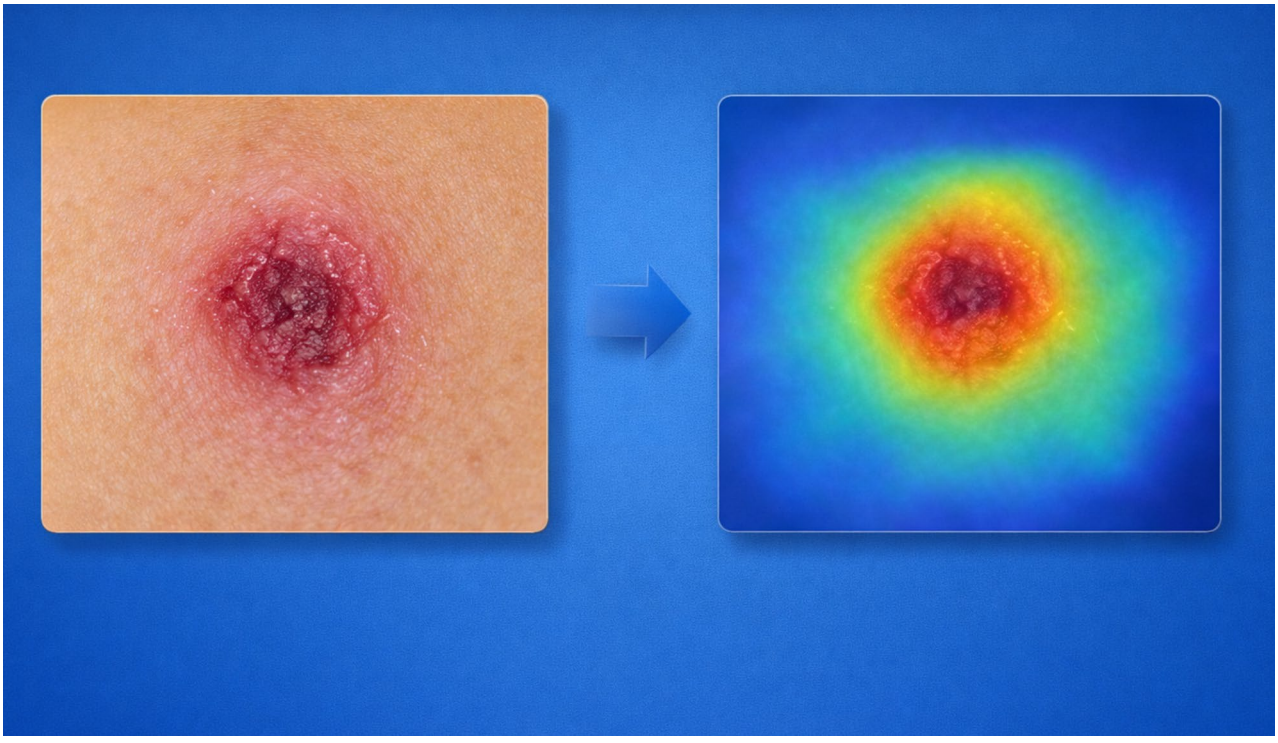


Fig. 2 Conceptual illustration of explainable artificial intelligence (XAI) output in skin disease diagnosis, showing a clinical skin lesion image alongside a corresponding attention-based heatmap highlighting regions that most strongly influenced the model's diagnostic prediction

Table 6 Differential diagnostic challenges in skin-manifesting NTDs and AI opportunities

NTD	Clinically Similar Conditions	Key Diagnostic Challenge	Risk if Misclassified	AI Opportunity
Scabies	Eczema, impetigo, insect bites	Non-specific pruritic lesions	Overtreatment or missed infestation	Pattern recognition of lesion distribution
Yaws	Tropical ulcers, bacterial skin infections	Similar ulcer morphology	Delayed eradication efforts	Temporal lesion progression analysis
Leprosy	Tinea, vitiligo, psoriasis	Subtle early lesions	Irreversible nerve damage	Texture and border detection
Cutaneous leishmaniasis	Buruli ulcer, skin cancer	Chronic painless ulcers	Delayed curative treatment	Ulcer edge and granulation mapping
Buruli ulcer	Leishmaniasis, pyoderma	Late presentation	Extensive tissue destruction	Early lesion evolution modelling

or disability risk [1]. Evidence from dermatology AI studies indicates that misclassification rates increase substantially when models are applied to under-represented populations or unfamiliar disease distributions, a concern that is amplified for skin NTDs affecting darker-skinned populations [22, 23].

In this context, explainable AI (XAI) functions as a critical safeguard in differential diagnosis rather than a purely interpretive enhancement. Visual explanations such as Grad-CAM heatmaps enable clinicians to assess whether model attention is focused on clinically meaningful lesion characteristics such as ulcer margins, scaling patterns or nodular architecture rather than irrelevant background features [25, 27]. By exposing potential failure modes and encouraging clinician–AI interaction, explainability supports safer deployment of AI systems in settings where diagnostic uncertainty and disease overlap are the norm rather than the exception (Table 6).

Practical considerations for LMIC deployment

One of the most significant challenges is image quality variability, driven by inconsistent lighting conditions, diverse skin tones, camera limitations and non-standardised image acquisition practices [12]. Studies have shown that reduced image resolution, motion blur and suboptimal illumination can substantially degrade AI diagnostic accuracy, particularly for subtle lesions common in early-stage NTDs [11, 16]. This variability necessitates robust preprocessing pipelines and quality-control mechanisms capable of flagging unreliable inputs prior to inference.

Equally critical is the requirement for offline functionality, as many endemic regions experience intermittent or absent internet connectivity. AI systems that rely exclusively on cloud-based inference risk limited usability and inequitable access, undermining their public health value [6, 7, 13, 14, 31, 32]. On-device or edge-based AI inference has therefore emerged as a pragmatic solution, enabling real-time decision support while preserving data privacy and operational continuity in remote settings [24]. However, offline deployment introduces computational constraints that must be balanced against model complexity and explainability requirements.

Device heterogeneity further complicates implementation, as frontline health workers use a wide range of smartphones with varying camera quality, processing power and operating systems. Evidence suggests that model performance may vary significantly across devices, introducing an additional layer of bias if not explicitly addressed during model development and validation [23]. Bias-aware evaluation across device classes is thus essential to ensure equitable performance.

Based on this, meaningful impact depends on integration into existing frontline workflows rather than parallel or disruptive systems. AI tools must align with clinical decision-making processes, support rather than replace health workers, and provide actionable outputs such as referral guidance or confidence indicators [12]. Without careful workflow integration and user-centred design, even technically robust AI systems risk poor adoption and limited sustainability in LMIC health systems.

Implications of this review

Previous studies have demonstrated that deep learning models can achieve dermatologist-level performance for selected skin conditions, largely within controlled datasets dominated by lighter skin tones and non-infectious diseases [15, 16]. Evidence of performance disparities across Fitzpatrick skin types IV–VI highlights persistent equity concerns in dermatological AI, particularly for populations with darker skin tones. More recent work has acknowledged performance disparities across skin tones and highlighted algorithmic bias as a structural limitation of current AI systems [22, 23]. This review advances the field by synthesising evidence that these limitations are particularly consequential for NTDs, where diagnostic uncertainty is already high and misclassification carries significant clinical and public health consequences [1, 12].

Clinical implications

From a clinical perspective, the integration of XAI into diagnostic workflows has the potential to substantially improve diagnostic confidence, particularly in differentiating visually similar skin NTDs. By making model reasoning transparent through heatmaps, confidence scores and attention mechanisms, clinicians can critically

appraise AI outputs rather than accept them uncritically, reducing over-reliance on automated systems [25]. For frontline health workers operating in low-resource settings, such tools can provide structured decision support, compensating for limited specialist access and supporting task-shifting without undermining clinical autonomy [12]. Importantly, improved diagnostic accuracy and transparency may reduce diagnostic delay and misclassification, which are known drivers of prolonged morbidity, disability and transmission in diseases such as leprosy and Buruli ulcer [1].

Public health and equity implications

At the population level, this review underscores the role of explainable AI in addressing diagnostic inequities affecting people with darker skin tones, who have historically been marginalised in both dermatological research and AI development [21, 22]. Equitable diagnostic tools are essential for achieving global NTD elimination targets, as under-diagnosis in specific populations undermines surveillance accuracy and intervention effectiveness [1]. Moreover, embedding transparency and fairness into AI systems aligns with emerging ethical frameworks for responsible digital health deployment in endemic regions [1, 24].

Research and policy implications

The review highlights an urgent need for inclusive, skin-tone-diverse datasets and routine performance stratification in AI research to prevent the perpetuation of hidden biases. It also reinforces the importance of explainability as a regulatory requirement, not merely a technical enhancement, as policymakers increasingly scrutinise AI systems used in clinical care [24]. Finally, explainable AI offers promising opportunities for future AI-driven NTD surveillance, enabling interpretable aggregation of image-based data to support early outbreak detection and targeted public health responses.

Limitations of this review

This review limitations include that first, the literature search was restricted to English-language publications, which may have excluded relevant studies published in other languages, particularly from endemic regions where skin-manifesting NTDs are prevalent. Also, the rapidly evolving nature of artificial intelligence research means that some recently developed models or preprints may not have been fully captured. Again, substantial heterogeneity in study design, datasets, AI architectures, and reporting standards limited direct quantitative comparison across studies. Finally, the limited availability of NTD-specific AI studies constrained the ability to draw disease-specific performance estimates, necessitating cautious interpretation of generalisability.

Conclusion

This review synthesised current evidence on the application of explainable artificial intelligence (XAI) for the differential diagnosis of skin-manifesting neglected tropical diseases (NTDs), with a specific focus on darker skin tones. The findings demonstrate that while convolutional neural networks and related deep learning models have achieved high diagnostic performance in dermatology often reporting accuracy or AUC values exceeding 85–90% in controlled settings substantial performance gaps persist across skin tones, with reductions of up to 15–20% reported in darker-skinned populations. These disparities are largely attributable to dataset imbalance, algorithmic bias, and limited evaluation stratified by skin tone. In addition, it further highlights that explainability techniques, including Grad-CAM heatmaps and confidence scoring, offer critical safeguards in differential diagnosis by enhancing transparency, clinician trust, and error detection. However, explainability alone is insufficient without inclusive data, context-aware model design, and integration into frontline workflows. For skin-manifesting NTDs where visual overlap, diagnostic delay, and misclassification are common XAI presents a uniquely valuable opportunity to improve diagnostic confidence, support non-specialist health workers, and advance equitable care. Future research must prioritise diverse datasets, standardised reporting, and regulatory frameworks that mandate explainability, ensuring that AI-driven diagnostics contribute meaningfully to NTD control and elimination efforts in low- and middle-income countries.

Abbreviations

AI	Artificial Intelligence
CDC	Center for Disease Control and Prevention
CI	Confidence Intervals
CNNs	Convolutional neural networks
LMICs	Low-and middle-income countries
MESH	Medical Subject Headings
ML	Machine Learning
NTDs	Neglected Tropical Diseases
RCT	Randomised control Trials
WHO	World Health Organization
XAI	Explainable Artificial Intelligence

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Competing interests
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